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TITLE PAGE

- **Problem Statement ID –25029**
- **Problem Statement Title-Authenticity Validator for Academia**
- **Theme- Smart Education**
- **PS Category- Software**
- **AISHE code : C-33505**
- **Team ID-34**
- **Team Name-TEAM INSTINCT**

TEAM INSTINCT

IDEA TITLE

A Blockchain-Based Solution for Certificate Authentication

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What it is : A digital platform powered by blockchain technology.

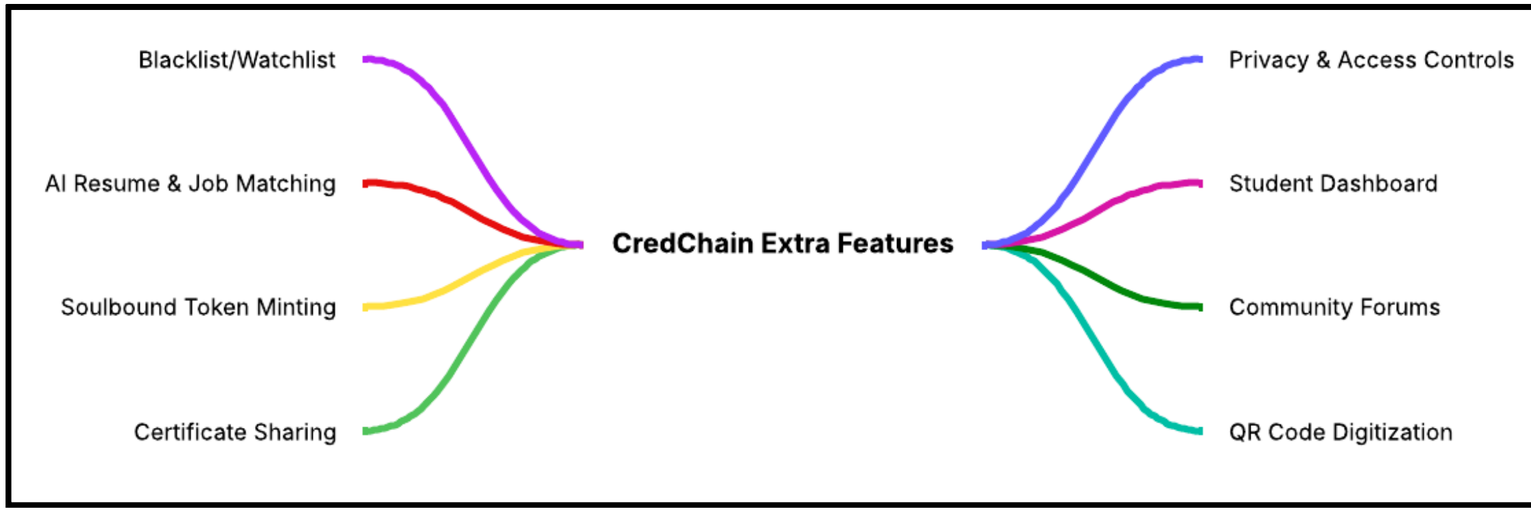
Main Purpose: To verify academic certificates and detect fake or forged degrees.

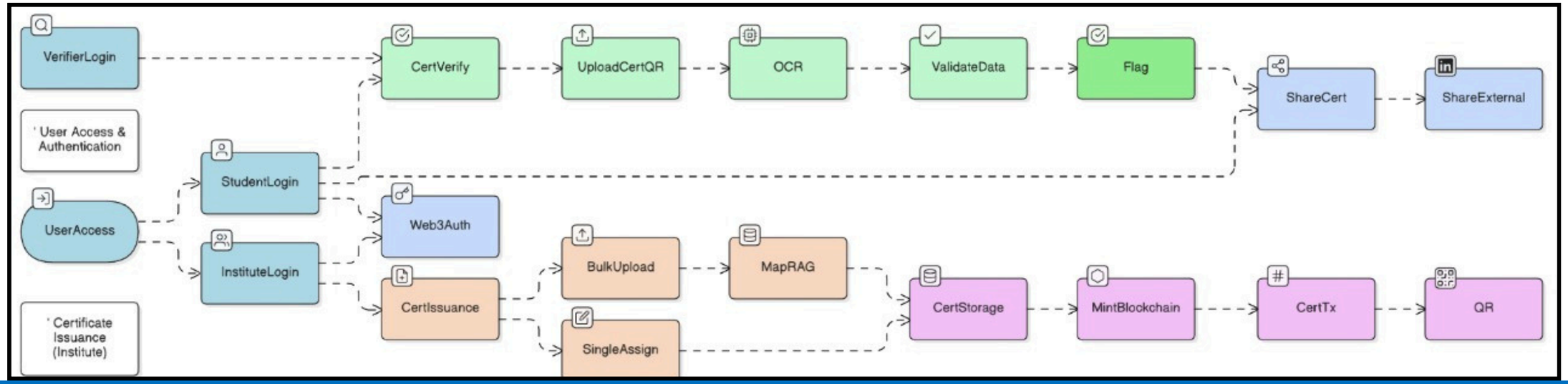
Why it's needed: Addresses the growing problem of fake academic credentials in higher education.

How it works:

- Uses blockchain to securely store and verify academic records.
- Uses a Merkle tree for cost-effective minting.
- Ensures that certificates are tamper-proof and cannot be faked.

Key Benefit: Creates a trustworthy system where academic credentials are permanently and securely validated.





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TECHNICAL APPROACH

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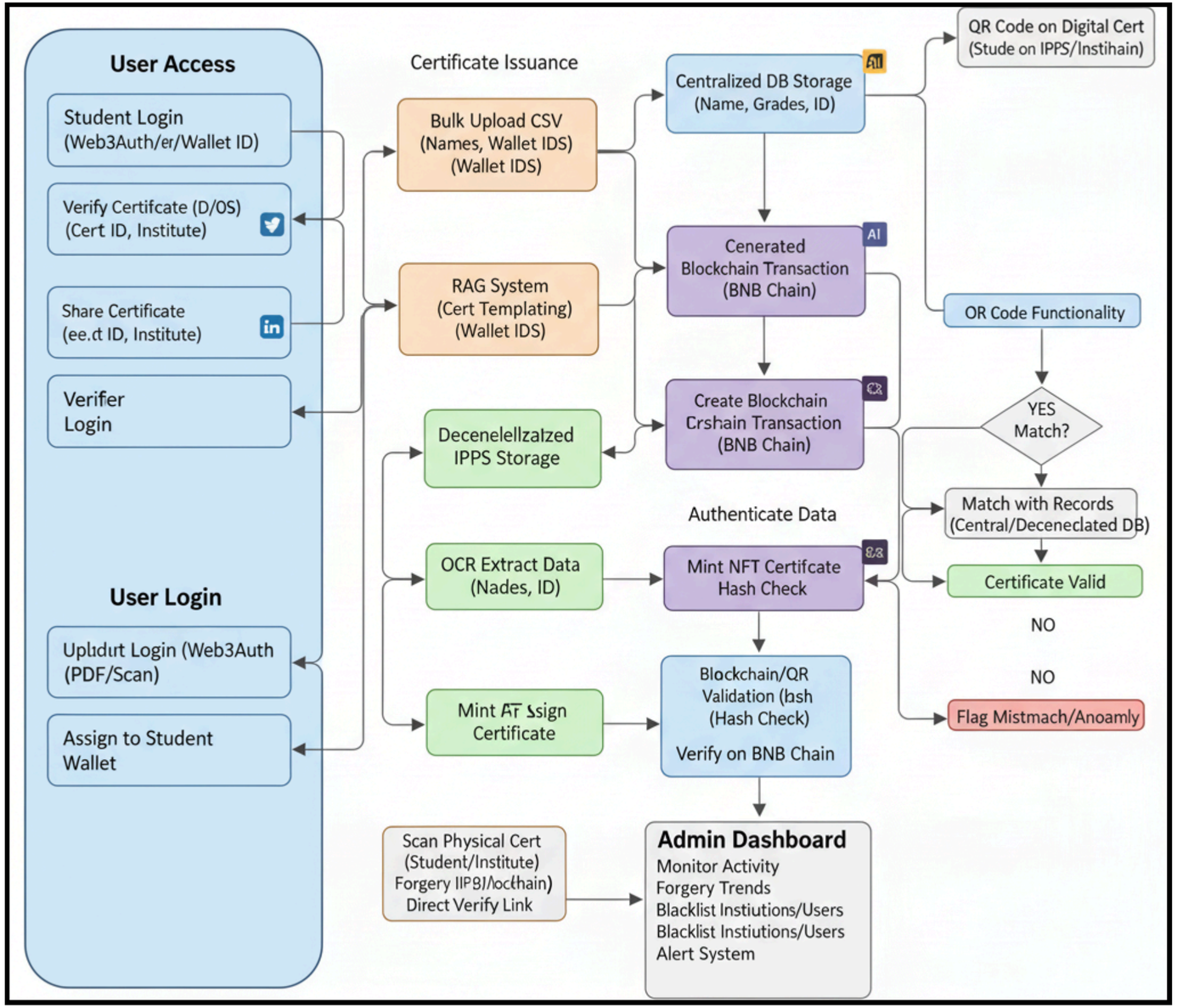


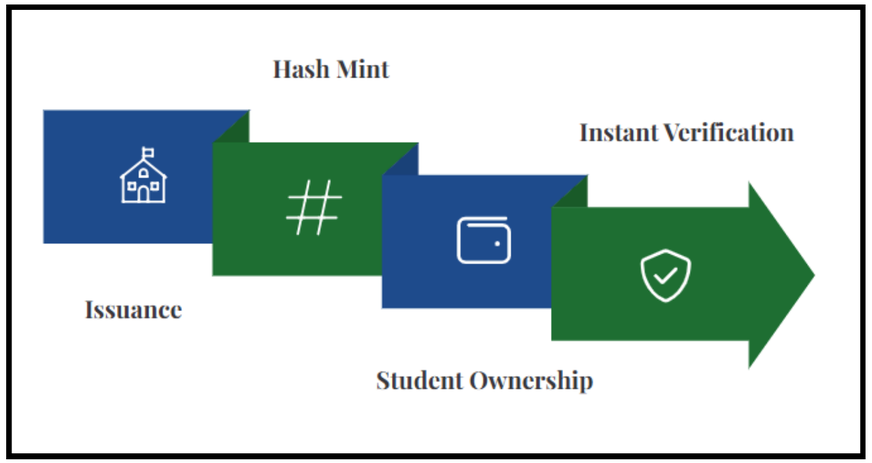
TECH STACK USED

FRONTEND: REACT, METAMASK, TAILWIND CSS

BACKEND: SOLIDITY, OPBNB

DATABASE/STORAGE: MONGODB

AUTHENTICATION: WEB3AUTH



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FEASIBILITY AND VIABILITY

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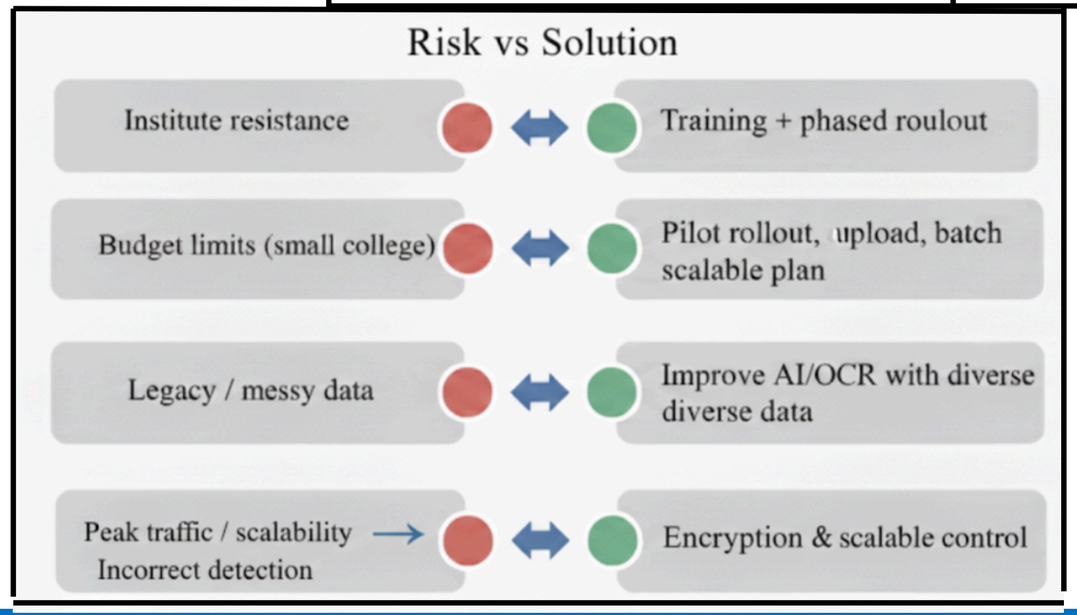
Aspect	Observation	Notes
Technical Feasibility	Blockchain + OCR	Easy to implement
Cost / Budget	Low deployment cost	Affordable
User Adoption	Web dashboards+Web3 auth login	Less tech-savvy users
Integration with Institutes	ERP + legacy support	Flexible uploads
Scalability	Many users & bulk uploads	Can grow statewide

OCR: Optical Character Recognition.
ERP:Enterprise Resource Planning.
SBT: Soulbound Token



Business Potential :

- **Cost efficiency** → Cuts admin & fraud costs
- **Revenue** → Subscription / API fees
- **Partnership ready** → Govt & Smart City programs
- **Scalable** → Statewide & multi-institute use
- **Global portability** → Valid for international hiring/studies
- **Fraud-proof** → Blockchain + SBT
- **AI-powered** → Automatic anomaly detection

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IMPACT AND BENEFITS

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Cost efficiency

Automated, paperless checks replace manual follow-ups and reduce per-verification and admin costs.

Immutable trust

Blockchain's append-only ledger makes records tamper-evident, boosting confidence and reducing certificate forgery.

Instant verification

QR/on-chain lookups confirm credentials in seconds, speeding admissions, offers, and onboarding.

Instant checks

QR scans and on-chain lookups confirm authenticity in seconds, slashing verification time and cost.

Assured authenticity

Cryptographic proofs and non-transferable credentials prevent tampering, resale, and misuse.

Bound identity

Non-transferable SBTs lock credentials to the rightful holder's wallet, preventing resale or hand-offs.

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RESEARCH AND REFERENCES

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All documents, references have been listed properly in the attached document with the QR. Kindly Scan to know more about the Methodology.

Link to the sources

Unique Feature	Credly	ChainCred
Non-transferable (Soulbound) credentials	❌	✅
Wallet-based login (Web3Auth, etc.)	❌	✅
AI-driven anomaly detection	❌	✅
OCR certificate information extraction	❌	✅
Physical certificate digitization via QR	❌	✅
AI-powered job recommendations	❌	✅
Real-time alerts on anomalies/invalidation	❌	✅